

AI-Powered Delivery Governance and KPI Dashboard

Business Requirements Document

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1 Executive Summary

The **AI-Powered Delivery Governance and KPI Dashboard** is an intelligent enterprise solution designed to provide centralized visibility into software delivery performance, project governance, sprint execution, quality metrics, productivity measurements, risk management, and executive reporting. The platform integrates with Azure DevOps and leverages Artificial Intelligence to automate delivery insights, improve decision-making, and reduce the manual effort associated with project monitoring and reporting.

The system continuously collects project, sprint, work item, and quality data from Azure DevOps through REST APIs and processes the information using a combination of KPI analytics, governance rules, and Azure OpenAI-powered intelligence engines. By analyzing delivery trends and performance patterns, the platform generates AI-driven insights, identifies risks, predicts delivery issues, and provides actionable recommendations to project managers, Scrum Masters, PMOs, and executive stakeholders. The platform offers interactive dashboards, automated KPI tracking, delivery health monitoring, sprint analytics, resource productivity measurements, and executive summaries through a centralized governance interface. AI-generated reports and management insights provide leadership teams with real-time visibility into project status, enabling faster and more informed decision-making.

By combining Azure DevOps, and Azure OpenAI capabilities, the proposed solution transforms traditional project reporting into an intelligent delivery governance ecosystem that enhances transparency, accountability, and operational excellence across software development teams.

2 Project Overview

2.1 Introduction

Modern organizations execute software delivery through Agile methodologies, utilizing platforms such as Azure DevOps to manage work items, user stories, tasks, defects, releases, and sprint activities. While these tools provide extensive delivery data, organizations often struggle to consolidate, analyze, and transform this information into meaningful insights that support governance and decision-making. Project managers and leadership teams frequently depend on manual reporting processes, spreadsheets, and fragmented dashboards to track delivery performance. This approach is time-consuming, error-prone, and often provides only retrospective visibility into project outcomes. Risks, delivery delays, quality concerns, and productivity issues are frequently identified too late, resulting in reactive rather than proactive management.

The proposed AI Delivery Governance Platform addresses these challenges by integrating directly with Azure DevOps, collecting delivery metrics in real time, and leveraging Artificial Intelligence to generate automated insights, executive summaries, KPI evaluations, risk assessments, and performance recommendations. The platform provides a unified governance framework that improves visibility, enhances accountability, and supports data-driven delivery management.

2.2 Project Vision

To develop an AI-powered Delivery Governance Platform that provides real-time visibility into project health, sprint execution, delivery performance, quality metrics, productivity measurements, and governance KPIs by integrating Azure DevOps data with Azure OpenAI-powered analytics and reporting capabilities. The platform aims to empower project teams, managers, PMOs, and executive leadership with intelligent insights, predictive risk analysis, and automated reporting to improve delivery success, reduce governance overhead, and enable proactive decision-making throughout the SDLC.

2.3 Project Objectives

- Develop a centralized delivery governance platform integrated with Azure DevOps.
- Track sprint execution, velocity, backlog health, and delivery progress.
- Monitor quality metrics including defects, bug trends, and test outcomes.
- Measure team productivity and engineering performance.
- Identify delivery risks using AI-powered analysis.
- Reduce manual reporting effort and governance overhead.

3 Business Problem Statement

Organizations increasingly rely on Agile delivery methodologies and Azure DevOps to manage software projects. These systems generate significant volumes of data related to project progress, sprint performance, work item completion, quality metrics, team productivity, and delivery timelines. Despite having access to large amounts of operational data, many organizations face significant challenges in transforming this information into actionable business insights. Delivery information is often scattered across multiple dashboards, reports, and project repositories, making it difficult for stakeholders to gain a comprehensive view of project performance.

Traditional reporting processes require substantial manual effort to consolidate delivery data, calculate KPIs, prepare executive reports, and assess risks. As a result, critical issues such as sprint delays, productivity decline, quality degradation, or schedule risks are frequently identified only after impacting project outcomes.

Additionally, existing dashboards primarily focus on historical metrics and lack the intelligence required to explain delivery trends, predict future risks, or recommend corrective actions. Leadership teams require deeper visibility into delivery performance without depending on manual analysis and status meetings.

The proposed **AI-Powered Delivery Governance and KPI Dashboard** addresses these challenges by providing centralized governance, automated KPI monitoring, AI-driven risk identification, delivery intelligence, predictive analytics, and executive reporting. The solution enables proactive delivery management while reducing manual reporting effort and improving organizational decision-making.

3.1 Business Benefits

The proposed solution provides the following business benefits:

- Real-time visibility into project and sprint execution.
- Automated KPI tracking and governance reporting.
- AI-powered delivery insights.
- Early identification of project risks and delivery bottlenecks.
- Improved decision-making through predictive analytics.
- Reduced manual effort in data consolidation and reporting.
- Centralized governance across projects and teams.
- Increased delivery predictability and accountability.
- Better alignment between project execution and business objectives.

4 Business Objectives

The following business objectives define the strategic outcomes expected from implementing the proposed solution.

ID	Business Objective
BO-01	Establish a centralized delivery governance platform integrated with Azure DevOps.
BO-02	Automate delivery KPI collection, monitoring, and reporting.
BO-03	Provide real-time visibility into sprint execution and project performance.
BO-04	Measure and analyze quality, productivity, and delivery metrics.
BO-05	Detect project risks and delivery issues using AI-powered analysis.
BO-06	Generate automated executive reports and delivery summaries.
BO-07	Enable data-driven decision-making through predictive analytics.
BO-08	Reduce governance overhead and manual reporting efforts.
BO-09	Provide secure role-based access through Microsoft Entra ID.
BO-10	Deliver a scalable and intelligent governance platform capable of supporting enterprise-wide delivery management.

5 Scope Definition

5.1 In Scope

The AI-Powered Delivery Governance and KPI Dashboard shall include the following capabilities:

- Integration with Azure DevOps using REST APIs.
- Synchronization of projects, epics, user stories, tasks, bugs, sprints, iterations, repositories, and pipelines.
- Centralized storage of synchronized data in Azure SQL Database.
- Delivery KPI calculation and monitoring.
- Sprint governance and delivery tracking.
- AI-powered risk analysis and predictive insights.
- AI-generated executive summaries and governance reports.
- Role-based dashboard visualization.
- Microsoft Entra ID authentication and authorization.
- Audit logging and synchronization monitoring.
- Historical trend analysis and reporting.

5.2 Out of Scope

The following functionalities are excluded from the MVP release:

- Project budgeting, financial management and Resource allocation.
- Mobile application development.
- Integrations with Jira, ServiceNow, GitHub, or other third-party systems.
- Employee performance appraisal management.

5.3 Assumptions

- Azure DevOps services are available and accessible.
- Required API permissions are granted.
- Users possess valid Microsoft Entra ID accounts.
- Azure OpenAI services are provisioned.
- Organizational governance policies permit delivery data analysis.

6 Stakeholder Management

Table 1: Key Stakeholders and Responsibilities

Stakeholder	Role	Responsibilities
Administrator	Organization Owner	Has organization-wide access to all projects and dashboards. Monitors project health, reviews AI insights, and oversees delivery governance.
Project Manager	Primary User	Tracks project progress, sprint performance, delivery health, team productivity, project risks, and KPI trends.
IT Support Team	Support User	Configures integrations, manages users, maintains the platform, and monitors application health and performance.

7 Business Rules

7.1 Delivery Metrics Rules

BR-01 Sprint Completion Rate shall be calculated as:

$$\text{Sprint Completion Rate} = \left(\frac{\text{Completed Story Points}}{\text{Planned Story Points}} \right) \times 100$$

BR-02 A sprint shall be classified as “**At Risk**” when Sprint Completion Rate falls below 80%.

BR-03 Defect Density shall be calculated as:

$$\text{Defect Density} = \frac{\text{Total Defects}}{\text{Completed Story Points}}$$

BR-04 Backlog Health shall be classified as:

- Healthy: > 85%
- Moderate: 70% – 85%
- Poor: < 70%

BR-05 A project shall be considered delayed when more than 20% of planned work remains incomplete after sprint closure.

7.2 Security Rules

BR-06 Only Administrators can access organization’s whole process.

BR-07 Project Manager can access only projects which are dealing by him.

BR-08 IT Admin can view Application’s logs and details.

7.3 Data Retention Rules

BR-10 Audit logs shall be retained for 90 days.

BR-11 KPI historical data shall be retained for a minimum of 24 months.

BR-12 Synchronization logs shall be retained for 12 months.

7.4 AI Risk Rules

BR-13 Risk scores shall be categorized as shown in Table 2.

Table 2: Risk Score Classification

Score Range	Risk Level
0–30	Low
31–60	Medium
61–80	High
81–100	Critical

BR-14 Projects classified as High or Critical risk shall automatically appear on governance dashboards.

8 Non-Functional Requirements

8.1 Performance Requirements

Table 3: Performance Requirements

ID	Requirement
NFR-01	The system shall load dashboard pages within 3 seconds for 95% of user requests.
NFR-02	The system shall generate executive reports within 30 seconds.
NFR-03	KPI calculations shall complete within 5 minutes after successful data synchronization.
NFR-04	Risk analysis results shall be generated within 60 seconds after AI processing is initiated.
NFR-05	Search and filtering operations shall return results within 2 seconds.

8.2 Security Requirements

Table 4: Security Requirements

ID	Requirement
NFR-06	All user authentication shall be performed through Microsoft Entra ID.
NFR-07	Sensitive data shall be encrypted at rest using Azure encryption services.
NFR-08	The system shall enforce role-based access control (RBAC) for all users.
NFR-09	Unauthorized access attempts shall be logged and monitored.
NFR-10	User sessions shall automatically expire after 30 minutes of inactivity.

8.3 Reliability Requirements

Table 5: Reliability Requirements

ID	Requirement
NFR-11	The system shall maintain data consistency between Azure DevOps and the governance repository.
NFR-12	Failed synchronization jobs shall be automatically retried up to three times.
NFR-13	Failed AI requests shall be logged and retried automatically where applicable.
NFR-14	The system shall prevent duplicate records during synchronization processes.
NFR-15	All critical processing failures shall generate administrative alerts.

8.4 Usability Requirements

Table 6: Usability Requirements

ID	Requirement
NFR-16	The user interface shall provide a consistent look and feel across all modules.
NFR-17	Dashboards shall be optimized for desktop and standard enterprise laptop displays.
NFR-18	Users shall be able to access key insights within three clicks from the dashboard home page.
NFR-19	Error messages shall be displayed in a clear and user-friendly manner.

8.5 Audit and Logging Requirements

Table 7: Audit and Logging Requirements

ID	Requirement
NFR-20	All user login, logout, and administrative activities shall be recorded in audit logs.
NFR-21	Synchronization activities shall be logged with timestamps and execution status.
NFR-22	AI-generated reports and risk assessments shall be traceable to their source data.
NFR-23	Audit logs shall be searchable by administrators.

8.6 Data Retention and Backup Requirements

Table 8: Data Retention and Backup Requirements

ID	Requirement
NFR-24	KPI historical data shall be retained for a minimum of 12 months.
NFR-25	Audit logs shall be retained for a minimum of 90 days.
NFR-26	Daily database backups shall be performed automatically.
NFR-27	Backup data shall be stored securely and encrypted.
NFR-28	Backup restoration procedures shall be tested periodically.

8.7 AI Governance Requirements

Table 9: AI Governance Requirements

ID	Requirement
NFR-29	AI-generated insights shall clearly indicate that recommendations are system-generated.
NFR-30	AI processing failures shall not impact dashboard availability.
NFR-31	AI-generated recommendations shall be traceable to supporting delivery metrics.
NFR-32	AI-generated executive summaries shall be based only on synchronized project data.
NFR-33	The system shall maintain logs of AI requests and responses for audit purposes.

9 Technology Stack

The proposed solution utilizes modern open-source technologies to build a scalable, secure, and AI-enabled web application for competency assessment, personalized learning, and certification management.

Category	Technology	Purpose
Frontend Framework	React.js	User Interface Development
Frontend Language	TypeScript	Frontend Application Development
Backend Language	C# (ASP.NET Core)	Backend Application Development
Database	Azure SQL Database	Data Storage & Analytics
Authentication	Microsoft Entra ID	Authentication & Authorization
AI Service	Azure OpenAI	Insight Generation & Risk Analysis
API Integration	Azure DevOps REST API	Project Data Retrieval
Version Control	Git & Gitlab	Source Code Management
IDE	Visual Studio 2026 / VS Code	Development Environment
Diagram Tool	Draw.io (diagrams.net)	Architecture and Workflow Diagrams
Documentation	LaTeX (Overleaf)	Project Documentation

Table 10: Technology Stack

10 AI Budget

10.1 Overview

The **AI-Powered Delivery Governance and KPI Dashboard** utilizes Azure OpenAI to generate delivery intelligence, executive summaries, sprint analysis, KPI explanations, productivity insights, and predictive risk assessments. AI capabilities significantly reduce manual governance activities while enabling proactive delivery management.

10.2 Estimated Production Cost

The platform follows a **pay-as-you-use** pricing model for AI services. The overall cost depends on the AI request frequency, selected AI model, and cloud provider. Table 11 presents an approximate estimate of AI processing costs.

Table 11: Estimated Monthly Azure Costs

Service	Monthly Cost (USD)
Azure OpenAI	\$5–\$20
Azure App Service (Basic)	\$10–\$20
Azure SQL Database (Basic)	\$5–\$15
Microsoft Entra ID	Free
Total	\$20–\$55 / month

Future Enhancement Note: The current cost estimation is based on the MVP implementation. Future releases may incorporate Agentic AI capabilities such as autonomous risk monitoring, automated corrective action recommendations. Such enhancements may increase AI resource consumption and associated cloud costs depending on the frequency of agent execution and workload volume.

10.3 Benefits of AI Integration

- Generates automated delivery insights and executive summaries.
- Performs intelligent KPI interpretation and trend analysis.
- Identifies delivery risks before project impact occurs.
- Detects sprint anomalies and performance bottlenecks.
- Provides predictive forecasting for delivery outcomes.
- Recommends corrective actions to improve project performance.
- Reduces manual reporting and governance effort.

11 Epic 1 - Azure DevOps Data Integration & Project Synchronization

This module serves as the foundation of the AI Delivery Governance Platform by integrating Azure DevOps and collecting project delivery data automatically. It retrieves project information, work items, sprint details, teams, repositories, pipelines, defects, and other delivery-related metrics required for governance and analytics.

11.1 Functional Requirements

ID	Requirement
FR-1.1	The system shall connect to Azure DevOps using REST APIs.
FR-1.2	The system shall authenticate using Azure DevOps Personal Access Token (PAT) or Service Principal.
FR-1.3	The system shall retrieve projects from configured Azure DevOps organizations.
FR-1.4	The system shall synchronize work items, user stories, tasks, bugs, and epics.
FR-1.5	The system shall retrieve sprint, iteration, and team information.
FR-1.6	The system shall store synchronized data in Azure SQL Database.
FR-1.7	The system shall support scheduled and manual synchronization.

11.2 Implementation Approach

The ASP.NET Core backend integrates with Azure DevOps REST APIs to retrieve delivery-related data. Data synchronization jobs execute periodically and fetch projects, work items, sprint details, bugs, repositories, and pipeline information. The retrieved data is transformed and stored within Azure SQL Database, creating a centralized governance repository. The synchronization engine maintains audit logs and supports incremental updates to minimize API consumption.

11.3 User Stories

- **US-1.1**

As a Administrator,

I want sprint, work item, repository, and pipeline data to be synchronized from Azure DevOps into the platform, So that project information is centrally available for governance and reporting.

- **US-1.2**

As a Project Manager,

I want Azure DevOps project data to be synchronized automatically at scheduled intervals, So that delivery metrics and project status information remain up to date.

- **US-1.3**

As an IT Admin,

I want to monitor synchronization jobs, application health, and integration status in real time, So that I can identify and resolve synchronization issues before they affect reporting and governance processes.

11.4 Acceptance Criteria

- Azure DevOps integration connects successfully.
- Project data is synchronized without duplication.
- Work items are imported correctly.
- Sprint data is available for analytics.
- Synchronization failures are logged.
- AI processing failures are logged and can be retried.
- Scheduled synchronization executes successfully.

12 Epic 2 - Authentication and Role-Based Access Control

This module provides secure access to the platform using Microsoft Entra ID and role-based authorization.

12.1 Functional Requirements

ID	Requirement
FR-2.1	The system shall authenticate users using Microsoft Entra ID.
FR-2.2	The system shall provide role-based authorization.
FR-2.3	The system shall support Executive, PMO, Manager, Scrum Master, and Admin roles.
FR-2.4	The system shall restrict access based on assigned roles.
FR-2.5	The system shall maintain user audit logs.
FR-2.6	The system shall prevent unauthorized access to protected REST API endpoints.

12.2 Implementation Approach

Microsoft Entra ID handles user authentication through OAuth 2.0 and OpenID Connect. User roles are mapped to platform permissions, ensuring secure access to dashboards, reports, and administrative functions.

12.3 User Stories

- **US-2.1**

As an IT Admin,

I want to securely log in to the application using my Microsoft Entra ID account, So that I can manage system configuration, monitor application health, and support users.

- **US-2.2**

As an Administrator,

I want to log in securely using my account, So that I can governance each project progress.

- **US-2.3**

As a Project Manager,

I want to securely log in and access only the projects assigned to me, So that I can monitor project delivery, KPIs, and project risks effectively.

12.4 Acceptance Criteria

- Users authenticate using Entra ID.
- Unauthorized access is prevented.
- User roles are correctly assigned.
- Access permissions function correctly.

13 Epic 3 - Delivery KPI Engine

This module calculates and tracks delivery KPIs across projects and teams.

13.1 Functional Requirements

ID	Requirement
FR-3.1	The system shall calculate sprint velocity.
FR-3.2	The system shall measure sprint completion rate.
FR-3.3	The system shall track defect density.
FR-3.4	The system shall calculate backlog health metrics.
FR-3.5	The system shall measure story completion trends.
FR-3.6	The system shall calculate release success rates.
FR-3.7	The system shall provide KPI trend analysis.

13.2 Implementation Approach

Data synchronized from Azure DevOps is processed by the KPI Engine to calculate governance metrics and performance indicators, which are stored for reporting and analysis.

13.3 User Stories

- **US-3.1**

As a Administrator,

I want to configure and monitor KPI calculation processes, So that delivery performance metrics are generated consistently across the organization.

- **US-3.2**

As a Project Manager,

I want to view delivery KPIs such as Sprint Completion Rate, Defect Density, and Backlog Health, So that I can track project performance and identify areas requiring attention.

13.4 Acceptance Criteria

- KPIs are calculated accurately.
- Historical KPI trends are available.
- KPI dashboards refresh automatically.

14 Epic 4 - Sprint Governance & Delivery Tracking

This module provides visibility into sprint execution and project delivery status.

14.1 Functional Requirements

ID	Requirement
FR-4.1	The system shall display sprint progress.
FR-4.2	The system shall track completed versus planned work.
FR-4.3	The system shall identify delayed work items.
FR-4.4	The system shall monitor sprint commitments.
FR-4.5	The system shall display burndown metrics.

14.2 Implementation Approach

Sprint-related information is processed to provide governance metrics, delivery status tracking, and sprint health indicators.

14.3 User Stories

- US-4.1

As an Administrator,

I want to view sprint governance metrics and delivery status across projects, So that I can identify delivery issues and ensure compliance with governance standards.

- US-4.2

As a Project Manager,

I want delayed work items, sprint risks, and delivery bottlenecks to be highlighted automatically, So that I can take corrective actions and improve sprint outcomes.

14.4 Acceptance Criteria

- Sprint progress is updated in real time.
- Burndown charts are displayed.
- Delayed items are highlighted.

15 Epic 5 - AI Risk Analysis Engine

This module leverages Azure OpenAI to identify and predict delivery risks.

15.1 Functional Requirements

ID	Requirement
FR-5.1	The system shall analyze delivery metrics using Azure OpenAI.
FR-5.2	The system shall identify delivery risks.
FR-5.3	The system shall predict schedule slippages.
FR-5.4	The system shall identify productivity bottlenecks.
FR-5.5	The system shall recommend mitigation actions.

15.2 Implementation Approach

The AI Engine sends project KPIs, sprint metrics, quality indicators, and trends to Azure OpenAI. The model evaluates delivery health and returns risk assessments and recommendations.

15.3 User Stories

- **US-5.1**

As a Project Manager,

I want AI-generated risk assessments and predictive insights for my projects, So that I can proactively identify potential delivery issues and take corrective actions before they impact project outcomes.

- **US-5.2**

As an Administrator,

I want AI-generated delivery forecasts and risk trends across projects, So that I can make informed governance decisions and address potential delivery challenges early.

15.4 Acceptance Criteria

- Risks are identified correctly.
- AI recommendations are generated.
- Risk explanations are available.

16 Epic 6 - AI Executive Reporting & Insights

This module generates executive summaries and governance reports.

16.1 Functional Requirements

ID	Requirement
FR-6.1	The system shall generate executive summaries.
FR-6.2	The system shall summarize project health.
FR-6.3	The system shall generate portfolio reports.
FR-6.4	The system shall create sprint review summaries.
FR-6.5	The system shall provide AI-generated recommendations.

16.2 Implementation Approach

Azure OpenAI transforms governance data into business-friendly summaries suitable for leadership and executives.

16.3 User Stories

- **US-6.1**

As an Administrator,

I want AI-generated executive reports and project summaries to be created automatically, So that I can review project performance, delivery metrics, and governance insights efficiently.

- **US-6.2**

As an Project Manager,

I want AI-generated project summaries and delivery insights, So that I can quickly understand project status, risks, and key performance indicators without manually analyzing large amounts of data.

16.4 Acceptance Criteria

- Executive summaries are generated.
- Reports are downloadable.

17 Epic 7 - Dashboards & Visualization

This module provides interactive governance dashboards for different stakeholders.

17.1 Functional Requirements

ID	Requirement
FR-7.1	The system shall provide role-based dashboards.
FR-7.2	The system shall display KPI metrics visually.
FR-7.3	The system shall display risk indicators.
FR-7.4	The system shall provide project health dashboards.
FR-7.5	The system shall support filtering and drill-down capabilities.

17.2 Implementation Approach

React.js dashboards consume APIs exposed by ASP.NET Core and present delivery, governance, and AI insights through charts and visual components.

17.3 User Stories

- **US-7.1**

As a Project Manager,

I want interactive delivery dashboards displaying project KPIs, sprint progress, and delivery health, So that I can monitor project performance and make timely decisions to improve delivery outcomes.

- **US-7.2**

As an Administrator,

I want portfolio-level dashboards and visualization of delivery trends across all projects, So that I can gain strategic insights and make informed governance decisions efficiently.

17.4 Acceptance Criteria

- Dashboards load successfully.
- KPIs are presented visually.
- Filtering works correctly.
- Dashboard data reflects current project status.

18 Project Timeline (1 Month)

Week	Activities
Week 1	Requirements gathering, Azure architecture design, database design, UI wireframes, Azure DevOps integration setup, project planning
Week 2	Backend API development, Azure DevOps synchronization module, authentication module, KPI engine implementation
Week 3	Sprint governance module, quality analytics module, Azure OpenAI integration, AI risk analysis, executive reporting.
Week 4	Dashboard development, notifications, testing, bug fixing, API validation, documentation, and final project deployment.

Table 19: Project Timeline

19 Conclusion

The **AI-Powered Delivery Governance and KPI Dashboard** provides a centralized and intelligent solution for monitoring software delivery performance, governance compliance, sprint execution, quality metrics, productivity analysis, risk management, and executive reporting. By integrating Azure DevOps, Azure SQL Database, Microsoft Entra ID, ASP.NET Core, React.js, and Azure OpenAI, the platform enables organizations to transform delivery data into actionable intelligence.

The solution automates KPI tracking, risk identification, executive reporting, and delivery analytics while reducing manual effort traditionally associated with governance and status reporting. Through AI-powered insights and predictive analysis, project teams, PMOs, managers, Scrum Masters, and executives can proactively address potential delivery issues and make informed decisions.

By combining real-time data integration, advanced analytics, intelligent reporting, and scalable cloud-native architecture, the AI Delivery Governance Platform establishes a modern governance framework that improves transparency, delivery predictability, operational efficiency, and overall project success.